



Low-Cost 40A, 1200V SOLID-STATE RELAY And VOLTAGE CONTROLLER

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Relays are essential components in electronic systems and are widely used in IoT and AC-powered applications to control loads using sensor outputs or low-voltage DC signals. Conventional electromechanical relays rely on a mechanical plunger that physically connects or disconnects AC contacts to perform switching.

By contrast, solid-state relays (SSRs) use semiconductor devices to perform switching, enabling faster, silent, and more efficient

operation without mechanical wear. An SSR allows current to flow through its control element when an appropriate input signal is applied. While electromechanical relays are limited to simple on/off operation, solid-state relays can also support voltage control and regulation functions. High-current solid-state relays, however, remain relatively uncommon and are typically restricted to a maximum current range of 1A to 10A in commercially available designs.

This system presents a fast-switching, voltage-controlled solid-state relay that surpasses mechanical relays in both switching speed and efficiency. Beyond basic on/off switching, the system supports voltage control, allowing the output voltage to be regulated as required.

Unlike conventional SSRs rated at 10A or 20A, this relay is designed to handle up to 40A of continuous current and to operate at up to

1200V. According to its datasheet, the BTA41-1200BW TRIAC has a 40A continuous current rating and a 1200V voltage rating. The circuit supports two operating modes: one functions as a simple AC power switch, while the other operates as a voltage-controlled switch for output voltage regulation.

The system employs a minimal set of components, resulting in a compact, straightforward implementation without compromising performance. Fig. 1 shows the prototype PCB used in this system. The components required to assemble the system are listed in the Bill of Materials table.

Circuit and working

Fig. 2 shows the circuit diagram of the solid-state relay and AC voltage controller. The circuit is built around optotriac MOC3041 (IC1), power TRIAC BTA41-1200BW (Q1), a DIAC (D1), and some discrete components.

Since the TRIAC handles high current, a suitable heat sink is mandatory for safe and reliable operation.

In this circuit, the selected TRIAC is the BTA41-1200BW. As per the manufacturer's datasheet (see Fig. 3), it can handle load currents of up to 40A and withstand voltages up to 1200V. The BTA41 series is available in several variants with different voltage and current ratings; readers are advised to refer to the datasheet for detailed options. In this system, the high-voltage, high-current variant

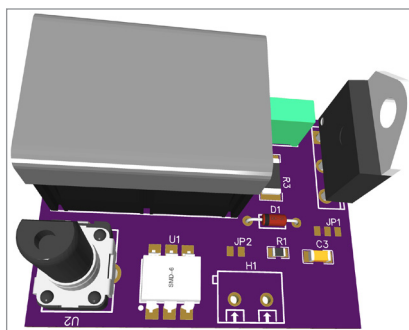


Fig. 1: Author's prototype PCB

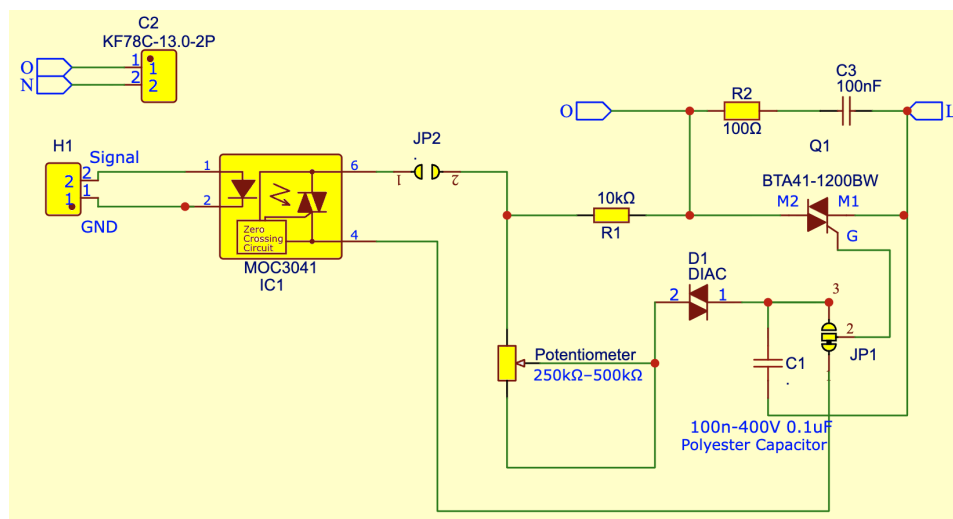


Fig. 2: Circuit diagram